

LP-relaxed Lagrangian relaxation

In the LP-relaxed Lagrangian relaxation approach, the demand coverage constraint is dualized and handled through a Lagrangian subgradient scheme.

In this variant, the subproblem associated with type-2 units P_2 is fully relaxed into a linear program by allowing maintenance decision variables to take fractional values :

$$x_{itk}, y_{it} \in \{0, 1\} \quad \rightarrow \quad x_{itk}, y_{it} \in [0, 1]$$

This relaxation transforms the maintenance subproblem into a pure LP, while production and stock variables remain continuous as in the original formulation.

Lagrangian relaxation handles coupling constraints by introducing Lagrange multipliers μ_t , which are updated using a subgradient method. This decomposes the problem into independent subproblems solved at each iteration.

The resulting dual problem is solved iteratively, and each evaluation of the dual function requires solving :

- a continuous production subproblem P_1 ,
- a relaxed maintenance-production subproblem P_2 (LP).

This yields fractional solutions for maintenance variables, which are not directly implementable but encode structural information about promising maintenance periods.

Method	P_2 type	Speed	Dual bound quality
MILP-based Lagrangian relaxation	MILP	slower	stronger
LP-relaxed Lagrangian relaxation	LP	faster	weaker

TABLE 1 – Trade-off between MILP and LP-relaxed Lagrangian approaches

The relaxed formulation of subproblem P_2 is given by :

$$\begin{aligned}
 (P_2) : \min \quad & \sum_{i \in I_2} \left[\sum_{k \in K_i} \left(RefCost_{ik} \sum_{t \in T} r_{it} \right) - \sum_{t \in T} \mu_t \cdot p_{it} \right] \\
 \text{s.c.} \quad & p_{it} \leq Pmax_{it}(1 - y_{it}) & \forall i \in I_2, \forall t \in T \\
 & s_{it} = s_{i,t-1} - p_{it} \times D_t + r_{it} & \forall i \in I_2, \forall t \in T \\
 & s_{it} \leq Smax_i & \forall i \in I_2, \forall t \in T \\
 & s_{it} \geq St h_i^{min} & \forall i \in I_2, \forall t \in T \\
 & r_{it} \leq Rmax_{ik} \times x_{itk} & \forall i \in I_2, \forall k \in K_i, \forall t \in T \\
 & \sum_{t \in k} x_{itk} \leq 1 & \forall i \in I_2, \forall k \in K_i \\
 & \sum_{t \in T} y_{it} = \sum_{k \in K_i} \sum_{t \in k} (Da_{ik} \times x_{itk}) & \forall i \in I_2 \\
 & \sum_{t' \in \{t, \dots, t+Da_{ik}\}} y_{it'} \geq Da_{ik} \times x_{itk} & \forall i \in I_2, \forall k \in K_i \\
 & x_{itk}, y_{it} \in [0, 1], p_{it} \geq 0, r_{it} \geq 0, s_{it} \geq 0
 \end{aligned}$$

The interpretation of this relaxed solution is fundamentally different from the original integer formulation. Fractional values of x_{itk} and y_{it} may appear, representing partially activated maintenance schedules. Although not physically implementable, these solutions provide useful dual information and guide the construction of feasible schedules.

Solution extraction : exact re-projection into a feasible schedule

Our implementation follows a deterministic three-stage pipeline :

1. Solve the Lagrangian dual problem using LP-relaxed subproblems P_1 and P_2 , optimized via a subgradient method ;
2. Construct an integer maintenance schedule using a deterministic projection operator based on the relaxed variables ;
3. Re-optimize production and fuel management with the maintenance schedule fixed, using a linear program.

The second stage does not rely on dual multipliers directly, but instead transforms the fractional maintenance solution into a feasible integer schedule.

Projection rule. For each maintenance campaign k of each type-2 unit i , the selected start time is defined by :

$$t^* = \arg \max_{t \in T_{ik}} x_{itk}$$

This rule selects the most activated start time in the relaxed solution.

Once t^* is chosen, the maintenance decision variables are reconstructed as :

$$x_{itk} = \begin{cases} 1 & \text{if } t = t^* \\ 0 & \text{otherwise} \end{cases}$$

The outage indicator is then expanded over the duration D_{aik} :

$$y_{it} = \begin{cases} 1 & \text{if } t \in [t^*, t^* + D_{aik}) \\ 0 & \text{otherwise} \end{cases}$$

This ensures temporal consistency between maintenance start decisions and induced outages.

Finally, once the maintenance schedule is fixed, production variables p_{it} , stock variables s_{it} , and refueling variables r_{it} are re-optimized through a linear programming model. Since all discrete decisions are fixed, this final step reduces to an LP.

Summary of the method

- LP relaxation of maintenance decisions in P_2 ,
- Lagrangian dual optimization via subgradient method,
- deterministic argmax-based projection of fractional solutions,
- LP re-optimization of production under fixed maintenance.

This approach produces a feasible solution to the original problem while preserving structural information from the relaxed dual solution.

Experimentations and results

Experimental Setup

The third version of our heuristic, *heuristique améliorée avec optimisation de x_{itk} et y_{it}* (presented in Section 3.3 of the main report), was re-executed on the same instances for comparison with the LP-relaxed Lagrangian relaxation approach. The MILP-based Lagrangian relaxation was not re-executed because its subproblem P_2 remains structurally close to the original problem. Consequently, under the 2-hour time limit considered in our study, its computational performance was expected to provide limited additional insight.

All experiments were conducted on a machine equipped with an 11th Gen Intel Core i5-1135G7 processor, 16 GB of RAM, running Ubuntu 24.04.4 LTS on a 64-bit Linux operating system.

Parameter Settings

The numerical results are reported in table 3. Regarding the subgradient algorithm parameters, the initial step size was set to 2, the reduction coefficient α to 0.95, the maximum number of iterations to 100, and the maximum execution time to 300 seconds.

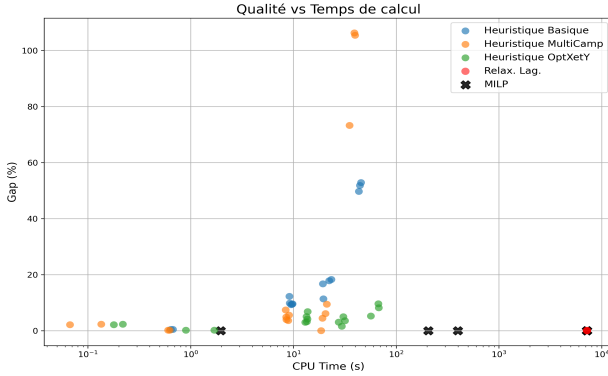
Numerical Results

Please refer to the main report for further details regarding GAP, CPU_2, and the remaining columns.

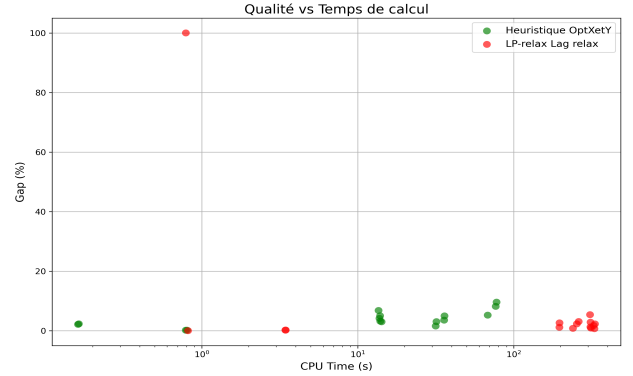
Method	Gap (%)			CPU_2 (s)			Taux de faisabilité (%)
	min	max	moyen	min	max	moyen	
Heur Opt. X et Y	0.16	9.59	3.94	0.16	77.67	26.833	100
LP-relax Lag. relax	0.00	5.38	1.69	0.82	332.03	225.73	94

TABLE 2 – PERFORMANCE SUMMARY

The results presented in 2 and 1 clearly show that the LP-relaxed Lagrangian relaxation achieves superior solution quality, with an average GAP of only 1.69% and a maximum GAP of 5.38%. However, this improvement comes at the expense of computational time, which remains higher than that of the slower heuristic version (Opt. X and Y). Despite this, the method continues to outperform full MILP resolution in speed while still delivering a high-quality approximate solution.



(a) Comparison between the three heuristic versions, Lagrangian relaxation, and the MILP model (same as in the main report ; ran on a different machine)



(b) Comparison between the best heuristic version and LP-relaxed Lagrangian relaxation (ran the machine described above)

FIGURE 1 – Trade-off between solution quality (Gap %) and computation time shown through scatter plots

Practical improvement

The quality of the solution can be improved by enhancing the subgradient scheme used in the dual optimization. In particular, adaptive step-size rules (e.g. diminishing step size or Polyak-type updates) and a bounded maximum number of iterations can stabilize convergence. Additionally, maintaining the best feasible projected solution across multiple dual iterations rather than only the final iterate can significantly improve robustness.

A Detailed results

Instances		Heuristique opt. de x_{itk} et y_{it}			LP-relaxed Lagrangian relaxation		
Data	S	VAL	Gap	CPU_2 (s)	VAL	Gap	CPU_2 (s)
toy	0	64624200	2.13%	0.161	Irréal. ¹	100%	0.791
toyy	0	60001800	2.30%	0.163	58651800	0.00%	0.817
data0	0	8624316633649.2	0.17%	0.805	8626907052831.601	0.20%	3.437
data0	1	8861032857649.2	0.16%	0.790	8865602917566	0.21%	3.456
data1	0	182070749047.46136	6.79%	13.604	174402452761.13492	2.29%	254.072
data1	1	172343616993.29266	4.14%	13.786	170599561826.9975	3.08%	261.090
data1	2	168049137082.36032	3.20%	13.951	164128742318.04712	0.79%	239.853
data1	3	155196821642.6063	2.99%	14.228	152412001116.75635	1.14%	196.351
data1	4	220637956147.5112	5.01%	13.916	215629243400.94183	2.63%	196.709
data2	0	151749111822.54932	3.04%	31.981	145985184284.1571	0.88%	312.987
data2	1	149743938404.81552	1.57%	31.613	143954303531.2663	5.38%	308.688
data3	0	146470880322.75027	4.97%	36.013	143510265738.30396	2.85%	311.034
data3	1	134996113638.58527	3.54%	35.814	132015833610.36536	1.25%	310.474
data4	0	126865240541.65536	9.59%	77.665	118390395577.05916	2.27%	332.029
data4	1	108804397056.5797	8.17%	76.692	101260728424.50978	0.67%	329.354
data5	0	119235885253.98442	5.22%	68.154	115288424007.98305	1.73%	325.592

TABLE 3 – Results of the LP-relaxed Lagrangian relaxation and the improved heuristic with x_{itk} and y_{it} optimization on our personal machine

1. To be noted that the toy instance itself is an artificial instance and it's feasible (shown in the main report). However, the LP-relaxed Lagrangian heuristic may return an infeasible schedule because the projection stage applies an independent argmax rule to fractional maintenance variables without enforcing global feasibility. This can generate overlapping outages that violate demand coverage. In addition, no feasibility-aware repair is performed during projection, and the reconstruction of x_{itk} may lose fractional information from the relaxed solution.